

Harshil Rawal

[LinkedIn](#) [GitHub](#) [LeetCode](#) [Codeforces](#)

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EDUCATION

- **Indian Institute of Technology, Mandi** Himachal Pradesh, India
B.Tech. in Mechanical Engineering — CGPA: 7.11/10 *August 2024 – 2028*
 - **Relevant Coursework:** Machine Learning, Probability & Statistics, Data Structures and Algorithms, Cybersecurity: Foundations, Systems, and Threat Analysis

EXPERIENCE

- **JPMorgan Chase & Co. – Quantitative Research Virtual Experience (Forage)** June 2026
Virtual Experience
 - Assessed credit risk by estimating Probability of Default (PD) from a loan portfolio using financial and credit data.
 - Implemented a dynamic programming algorithm to optimally bucket FICO scores for improved default prediction.
 - Used Python, NumPy, and Pandas for data preprocessing, feature engineering, and quantitative analysis.

TECHNICAL SKILLS

- **Languages:** C++, Python, JavaScript
- **Frameworks & Libraries:** Flask, React, PyQt6, SpeechRecognition
- **Data & ML:** NumPy, Pandas, Scikit-learn, Matplotlib, SQL
- **Tools:** Git, Linux, VS Code
- **Concepts:** Data Structures & Algorithms, OOP, Log Parsing, Regex, Credit Risk Modelling

PROJECTS

- **Linux Security Intelligence & Intrusion Detection System** [GitHub](#)
Python, Flask, Regex, Pandas, Matplotlib *May 2026*
 - Built a Linux intrusion detection system processing **10,000+ log entries** to detect brute-force login attempts, authentication failures, and suspicious IP activity using regex-based log parsing.
 - Developed a Flask dashboard automating threat visualization and generating structured TXT, JSON, and CSV reports for incident response workflows.
- **VoiceCalc: AI-Powered Speech Recognition Calculator** [GitHub](#)
Python, PyQt6, SpeechRecognition, SymPy, pyttsx3 *June 2026*
 - Developed a desktop scientific calculator supporting **40+ mathematical operations** through both voice commands and manual input using SpeechRecognition and SymPy.
 - Designed an intelligent speech parser for natural language math expressions, enabling real-time voice-to-computation conversion via a PyQt6 GUI with robust error handling.

ACHIEVEMENTS

- **Patent Applicant:** Led design of a Menstrual Waste Management System with IoT prototype (Patent: 202611043904).
- Achieved max rating of 1490 on LeetCode and 1000+ on Codeforces; solved 400+ DSA problems across platforms.

POSITIONS OF RESPONSIBILITY

- **Team Lead:** Design Practicum – led mechanical design and prototyping of the structure.
- **Publicity Volunteer:** Xpecto – drove 500+ registrations.
- **Core Member:** Dramatics Club, IIT Mandi – secured 6th rank in mime competition.